A Microscale Implementation of Ion Field Emitters in Single-Molecule Mass Spectrometry

Background and Motivation

Understanding how cells function at a molecular level requires a complete picture of the proteins that carry out essential processes, as proteins are the primary workforce of the cell; they take several important roles to catalyze reactions, regulate signaling networks, maintain structure, and respond dynamically to internal and external stimuli. As such, to truly understand cellular behavior, we must be able to detect and characterize proteins across the full range of their abundances and forms.

Mass spectrometry (MS) is the dominant technology for proteomics (the large-scale study and analysis of proteins in a cell or organism), as it allows us to measure molecular masses with high specificity and sensitivity. MS requires the molecules under investigation to be gas phase ions in vacuum, such that it can be put through a mass spectrometry process that selects and measures abundances of molecules with a specified charge to mass ratio. This is accomplished by a Nobel-prize winning method called electrospray ionization (ESI). In ESI, a protein solution is subjected to an electric field, which pushes droplets of a solution (a biological sample containing charged proteins, for example) out of a microscale nozzle into vacuum. It then enters a series of differential vacuum chambers, where charged droplets undergo successive Coulomb fissions. They split apart into smaller droplets as solvent evaporates, ultimately releasing ionized protein molecules that can be analyzed by their mass-to-charge ratios.[1]

However, ESI-MS faces two challenges:

First, its sensitivity is limited by its reliance on bulk ion generation and transfer. Only a small fraction of the ions generated are ultimately transmitted into the mass analyzer, and the process of droplet evaporation and Coulomb fission disperses ions, leading to inefficient collection of ions. This inefficiency makes it difficult to detect proteins that exist in very low numbers. In biological samples, such as a cell cytosol sample, the number of protein molecules per species can span many orders of magnitude, from just a handful of copies up to tens of millions per cell, where low-abundance proteins often play critical roles in regulation and signaling.[2]

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[1] https://www.mcponline.org/article/S1535-9476%2820%2930199-7
[2] https://pmc.ncbi.nlm.nih.gov/articles/PMC10044470

Second, ESI-MS typically requires proteins to be digested into peptides before analysis, and infers protein identity from peptide fragments. We lose information about the exact structure of the protein that was found in the sample, which is an issue due to how many proteins can exist in multiple proteoforms: distinct protein forms originating from the same parental gene arising due to various environmental factors within the cell. Each proteoform can have unique functions, and conventional bulk methods often cannot distinguish them or may alter their native states during preparation.

These limitations mean that deep proteome analysis (especially at the level of rare proteins and intact proteoforms) remains out of reach for most biological systems. This constrains our ability to investigate certain regulatory and signaling mechanisms, molecular events that may drive disease progression, developmental decisions, or responses to medication[3].

(IMAGE OF ESI vs IFE)

To address these challenges, our proposal focuses on the development of a robust ion source capable of single-molecule sensitivity. Specifically, we propose to use an **ion field emission (IFE)** nanopore method that applies an electric field to a protein solution through a nanoscale (<100 nm diameter) pore to directly eject individual protein molecules from the liquid phase into vacuum. This technique would bypass the droplet formation and turbulent ion generation inherent to ESI, producing a more collimated and efficient beam of ions that will be sent towards a micro-electrostatic trap (μEST), which acts as a high-sensitivity mass analyzer. By enhancing ionization and transmission efficiency at the source, this single-molecule approach has the potential to dramatically improve the detection of low-abundance proteins and intact proteoforms, allowing for deeper investigation into cellular function at a new level of precision. This has powerful applications in not just biological research, but also in medicine and biotechnology. For example, distinguishing variations in protein expression between cells in a tumor microenvironment can give insight into cancer cell behavior and treatment.[4]

The focus of this proposal is the design, simulation, and prototyping of the ion field emission nanopore source, laying the theoretical and experimental groundwork for single-molecule mass spectrometry with next-level sensitivity and specificity.

This project was recently approved as a collaboration between the Roukes group at Caltech and Stein group at Brown. The Stein Group was able to successfully create a ion emitter capable of emitting ions from ion solutions directly into vacuum.[5] Over 93% of the generated ions were

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[3] https://pmc.ncbi.nlm.nih.gov/articles/PMC8383168/#:~:text=The%20commonly%2Dused%20classical%20and,omics%20analyses%20of%20cell%20diversity.

[4] https://pubmed.ncbi.nlm.nih.gov/38359819/

[5] 1 Drachman, Nicholas, Mathilde Lepoitevin, Hannah Szapary, Benjamin Wiener, William Maulbetsch, and Derek Stein. "Nanopore ion sources deliver individual ions of amino acids and peptides directly into high vacuum." Nature Communications 15, no. 1 (2024): 7709.

successfully transmitted to a target Faraday cup through a 0.5m vacuum. However, this work has only been done through hand-assembled prototypes. In this project, we will design, manufacture, and test the first micro/nanofabricated IFE devices.

Objectives

What do you aim to accomplish in your project? What will you measure, and under what conditions; or, what will you calculate, model, or simulate? What are your starting assumptions or conditions, and what will be the result or product of a successful research project? What are the criteria for success? (In other words, how will you know when you have accomplished what you set out to do?)

The primary objective of this project is to develop and prototype a nanopore-based ion field emission (IFE) source capable of generating and directing ion beams suitable for integration with a single-molecule micro-electrostatic trap (μEST) mass analyzer.

Specifically, the objectives of this project are to:

1. Develop a physics-based simulation model describing ion emission and beam propagation from a nanopore IFE source through ion lenses as a function of applied electric potentials, nanopore geometry, and operating conditions.
2. Validate the model through comparison with experimental ion beam measurements using simple salts and model amino acids.
3. Design and prototype the mechanical and electrical framework required to align, focus, and condition the emitted ion beam for compatibility with downstream detection systems.

Here, we assume that ion evaporation from a nanopore emitter will produce an ion beam whose angular and velocity distributions are determined primarily by the local electric field geometry and operating potentials. By modeling and experimentally validating these dependencies, we aim to predict and control ion beam behavior.

The expected result of a successful project is then:

- A validated simulation tool capable of predicting ion beam behavior under given conditions
- A fabricated and operational IFE prototype with controllable emission properties
- A beam-conditioning and alignment framework capable of directing ions toward a micro-electrostatic trap

Methodology

Specifically, how will you accomplish your objective? What will you do? What are the principal steps or milestones along the path? How long will each take? What steps promise to be the most difficult, and how will you overcome the difficulties? What equipment or other resources will you need? Which of these are inherited, and which will you have to make or procure? With what other people or groups will you be collaborating? Will completion of your project depend on results from other people in related tasks? (That question may be especially pertinent for team projects.)

This project will have roughly three primary phases: (1) theoretical modeling and simulation of ion field emission and beam dynamics, (2) fabrication and prototyping of the nanopore IFE source and beam-conditioning interface, and (3) experimental validation and initial measurements.

Phase I: Modeling and Simulation of Ion Emission and Beam Propagation

The first phase of the project will focus on constructing a predictive simulation model of ion emission and beam trajectories from a nanopore ion field emission (IFE) source. Using COMSOL Multiphysics, we will simulate the electrostatic field geometry around the nanopore tip and calculate resulting ion trajectories as functions of applied extraction potentials, nanopore geometry, and operating temperature.

The goal of this phase is to determine how these variables influence:

- Ion velocity distributions
- Angular divergence of the beam
- Spatial beam profile downstream of the emitter

Initial assumptions will be informed by ion evaporation physics and prior work from the Stein group. Boundary conditions will employ information about vacuum and electrode configurations. The simulation will also then be used to predict how the ion beams can be directed and focused to a point (where the µEST mass analyzer will be) using Einzel lenses and deceleration plates.

This phase is expected to be the most challenging: to achieve agreement between simulation and physical behavior. The main difficulty would be ensuring that the physical assumptions embedded in the model accurately represent ion emission dynamics. To address this, the modeling process will be iterative: simulation predictions will be compared against experimental measurements (Phase III), and discrepancies will be resolved by refining assumptions/boundary conditions and consulting with my mentor and theoretical scientists within the group.

Phase II: Nanopore Fabrication and Interface Design

Based on simulation outcomes, optimized nanopore geometries and electrode configurations will be selected for fabrication. The actual nanofabrication of the IFE nozzles will be carried out at the Kavli Nanoscience Institute by a senior scientist in the group. Design discussions may also involve collaboration with the Stein group at Brown University, given their prior experience with nanopore IFE systems.

Concurrently, mechanical and electrical infrastructure for the emitter will be designed using CAD software such as SolidWorks. This will include:

- A precision mounting and alignment system capable of sub-micrometer positioning of the nanotip
- Structural components to hold electrostatic lens elements
- Points for high-voltage electrical connections
- Mounting for electrostatic deceleration plates

Electrostatic lens plates will be designed to focus the emitted ion beam, while deceleration stages will reduce ion kinetic energy to levels compatible with the micro-electrostatic trap (μEST). The μEST sensors themselves will be developed by other members of the laboratory, and coordination will ensure geometric and electrical compatibility between the ion source and the detector.

This phase will require access to laboratory tools such as, vacuum hardware, high-voltage power supplies, CAD software, and nanofabrication facilities. Many of these resources are available within the research group and institute. Custom components will be designed and fabricated as needed at the machine shop, as I am planning to take ME 13 over the summer to get machine-shop trained.

Phase III: Experimental Validation and Initial Measurements

After initial fabrication and assembly, the IFE source will be tested experimentally. Ion beam angular and spatial distributions will be measured using a downstream detector and compared to simulation predictions.

Simple salts and model amino acids (glycine, arginine, histidine, and tyrosine with m/z ranging from 75–181) will be used for validation. These molecules provide well-characterized reference masses for evaluating emission stability and beam control.

The primary goals of this phase are:

- To validate simulation predictions of beam angular and velocity distributions
- To assess focusing and deceleration performance
- To demonstrate controlled ion delivery consistent with requirements for single-molecule detection

If time allows and detector integration is available towards the end, initial measurements of analyte mass signals may be attempted.

Collaboration and Dependencies

This project is part of a larger effort to develop single-molecule mass spectrometry. While the μEST sensors will be developed by other members of the laboratory, this work focuses specifically on the ion source and beam-conditioning parts. Successful integration will require coordination with those team members to ensure alignment tolerances, voltage compatibility, and geometric constraints are met. Although integration with the μEST depends partially on progress by other team members, the modeling, fabrication, and beam characterization of the IFE source can proceed independently and will provide valuable standalone results even prior to full system integration.

In addition, consultation with the Stein group at Brown University will likely inform nanopore design decisions, leveraging their prior expertise in ion field emission systems.

Timeline

Making a schedule of the principal activities and events is a good way of showing the readers that you have taken a systematic approach to planning your work.

Weeks 1-4: Develop theory and set up COSMOL simulation of IFE nanopore and ion beam, complete first potential model and send in prototype nanopore design for nanofabrication
Weeks 5-6: Begin CAD, machining, and construction of spectrometer hardware
Weeks 7-8: Assemble hardware and set up electronics
Weeks 9-10: Assuming nanopore nozzles are manufactured and ready for use, implement in hardware and test, iterate on computational model and optimize parameters for highest transmission and sensitivity
Weeks 9-10: (If time allows, and assuming μEST mass analyzers are implemented) take initial data with mass spectrometer to test performance

References

List all pertinent papers or reports that you have consulted to prepare your proposal. Include remarks or suggestions from your prospective supervisor, from graduate students, or from other people with whom you have talked.